IIT HYDERABAD

Department of Electrical Engineering

Electronic Devices and Circuits Laboratory Report

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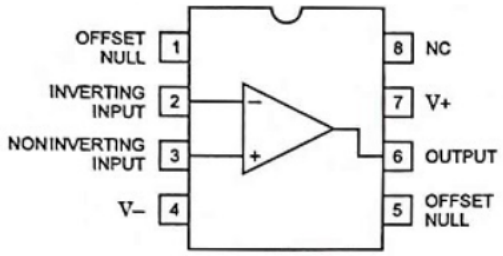
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1 Objective

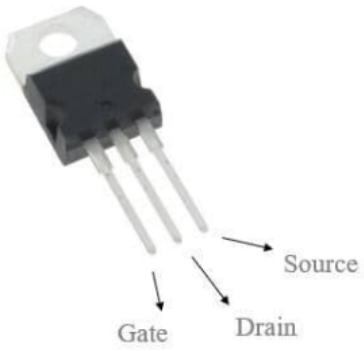
- To build a linear voltage regulator in which a 741 op-amp drives the gate of an NMOS power transistor connected as a source follower, so that the output is held at 1.2 V while the input supply varies from 3 V to 5 V.
- To measure line regulation, load regulation and dropout voltage, and to explain how negative feedback makes the output follow the reference. To measure the efficiency of the regulator.

2 Components and Equipment Required

1. 741 Op-Amp
2. Power N-Mosfet



(a) 741 Op-Amp



(b) IRF540N Power Mosfet

3. Output Capacitor ($C_{out} = 10\,\mu F$)
4. Resistors
   - Gate-Source Resistor $100\,k\Omega$
   - Bleeder Load $1\,k\Omega$
   - Load Resistors ($R_L = 124.9\,\Omega$, $46.8\,\Omega$, $22.4\,\Omega$, $12.2\,\Omega$)
5. DC Power Supply ($V_{AUX} = +12V, -12V$, $V_{IN} = 3.0V$ to $5.0V$ is given through it)
6. Function Generator ($V_{REF} = 1.2V$ is given through it)
7. Oscilloscope
8. Breadboard , Jumper wires

3 Theoretical Background

3.1 Circuit Description and Operating Principle

The NMOS linear regulator uses a 741 op-amp and a power MOSFET in negative feedback. The op-amp compares V_{REF} with V_{OUT} and adjusts the gate voltage to maintain regulation. The circuit includes $\pm 12\,\mathrm{V}$ supplies, a $100\,\mathrm{k}\Omega$ gate-to-source resistor, a $1\,\mathrm{k}\Omega$ bleeder resistor, and a $10\,\mu\mathrm{F}$ output capacitor. When V_{OUT} decreases, the op-amp increases the gate drive, restoring the output toward the reference voltage.

3.2 Output Voltage and Op-Amp Gain

For an op-amp with open-loop gain A and input offset voltage V_{OS}, the output voltage is approximately related to the reference voltage by

$$V_{\mathrm{OUT}} = \frac{A(V_{\mathrm{REF}} + V_{\mathrm{OS}}) - V_{\mathrm{GS}}}{1 + A}. \tag{1}$$

For an ideal op-amp with infinite gain and zero offset voltage, this reduces to

$$V_{\mathrm{OUT}} \approx V_{\mathrm{REF}}. \tag{2}$$

In practice, the output may differ slightly from the reference because of op-amp offset, reference variations, and measurement uncertainty. The gate voltage is related to the output by

$$V_G = V_{\mathrm{OUT}} + V_{\mathrm{GS}}. \tag{3}$$

3.3 MOSFET Characteristics

When operating in saturation, the MOSFET drain current can be approximated by

$$I_D = K(V_{\mathrm{GS}} - V_{\mathrm{TH}})^2(1 + \lambda V_{\mathrm{DS}}), \tag{4}$$

where K is the device transconductance parameter, V_{TH} is the threshold voltage, and λ represents channel-length modulation. The small-signal parameters are

$$g_m \approx \frac{2I_D}{V_{\mathrm{GS}} - V_{\mathrm{TH}}}, \qquad r_o \approx \frac{1}{\lambda I_D}. \tag{5}$$

As the load current increases, the gate voltage generally increases to supply the required current. When the input voltage approaches the output voltage, the MOSFET enters the triode region, and its limited on-resistance contributes to the dropout voltage.

3.4 Line and Load Regulation

Line regulation describes the change in output voltage with input voltage:

$$\text{Line Regulation} = \frac{V_{\text{OUT,max}} - V_{\text{OUT,min}}}{V_{\text{IN,max}} - V_{\text{IN,min}}} \quad (\text{mV/V}). \tag{6}$$

Load regulation describes the change in output voltage with load current:

$$\text{Load Regulation} = \frac{|\Delta V_{\text{OUT}}|}{|\Delta I_{\text{LOAD}}|} \quad (\text{mV/A}). \tag{7}$$

Negative feedback reduces the effective output resistance, allowing the regulator to maintain a nearly constant output despite changes in input voltage and load current. The total load current, including the permanent bleeder resistor, is

$$I_{\text{LOAD}} = \frac{V_{\text{OUT}}}{R_L} + \frac{V_{\text{OUT}}}{1\,\text{k}\Omega}. \tag{8}$$

3.5 Dropout Voltage

As V_{IN} decreases towards V_{OUT}, the MOSFET requires a smaller drain-to-source voltage to maintain the output. Eventually, the transistor can no longer regulate the output, which begins to decrease. The approximate voltage drop across the MOSFET is

$$V_{\text{IN}} - V_{\text{OUT}} \approx I_{\text{LOAD}} R_{\text{DS(on)}}. \tag{9}$$

In this experiment, the dropout threshold is defined as the input voltage at which the output falls by 1% below its regulated value.

3.6 Efficiency and Power Dissipation

The input power, output power, and power dissipated in the pass transistor are

$$P_{\text{IN}} = V_{\text{IN}} I_{\text{IN}}, \qquad P_{\text{OUT}} = V_{\text{OUT}} I_{\text{LOAD}}, \tag{10}$$

$$P_{\text{diss}} = P_{\text{IN}} - P_{\text{OUT}}. \tag{11}$$

The pass-stage efficiency is

$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% \approx \frac{V_{\text{OUT}}}{V_{\text{IN}}} \times 100\%. \tag{12}$$

Since the pass transistor operates in its linear region, the excess input power is dissipated as heat. Consequently, efficiency generally decreases as the input voltage increases

while the output remains constant. The auxiliary power consumed by the 741 can also be included when calculating the overall efficiency.

3.7 Stability

The output capacitor helps reduce voltage fluctuations and supports the stability of the feedback loop. However, the op-amp and MOSFET introduce frequency-dependent effects that may cause oscillations. Therefore, the output waveform is observed using an oscilloscope to examine ripple, transient fluctuations, and any sustained oscillations.

3.8 Comparison with a Switching Regulator

Unlike a linear regulator, a buck converter operates its switching transistor mainly in the fully ON or OFF state. For an ideal buck converter operating in continuous conduction,

$$V_{\mathrm{OUT}} = DV_{\mathrm{IN}}, \qquad D = \frac{t_{\mathrm{ON}}}{T}, \tag{13}$$

where D is the duty cycle. Switching regulators generally provide higher efficiency, whereas linear regulators offer simpler circuits and low output noise, making them suitable for low-noise applications.

4 Part A - Setup and Initial Verification

4.1 Experimental Procedure

The circuit was assembled as shown in the circuit diagram, with the bleeder resistor as the only load. The $+12\,\mathrm{V}$ and $-12\,\mathrm{V}$ supplies were switched on, and the reference voltage was adjusted to $V_{\mathrm{REF}} = 1.20\,\mathrm{V}$. The input voltage was then set to $V_{\mathrm{IN}} = 3.0\,\mathrm{V}$.

The output voltage V_{OUT} and the MOSFET gate voltage V_G were measured using a multimeter to verify the operation of the regulator.

4.2 Observations

The following voltages were measured during the initial verification of the regulator.

Table 1: Initial verification of the NMOS voltage regulator

Parameter	Measured Value
Reference voltage, V_{REF}	1.200 V
Input voltage, V_{IN}	3.0 V
Output voltage, V_{OUT}	1.1945 V
Gate voltage, V_G	3.908 V
Gate-to-source voltage, V_{GS}	2.718 V

4.3 Results and Discussion

The measured output voltage was 1.1945 V, which is close to the reference voltage of 1.200 V. The measured gate voltage was 3.908 V, and the gate-to-source voltage was 2.718 V.

The gate-to-source voltage is related to the gate and source voltages by

$$V_{GS} = V_G - V_{\text{OUT}}.$$

Using the measured gate and output voltages,

$$V_{GS} = 3.908 - 1.1945 = 2.7135\,\text{V}.$$

This is close to the directly measured value of 2.718 V, with a difference of 4.5 mV attributable to instrument accuracy limitations, measurement resolution and other experimental errors.

5 Part B - Line Regulation

5.1 Experimental Procedure

A load resistor of $R_L = 22.4\,\Omega$ was connected in parallel with the bleeder resistor. The input voltage V_{IN} was varied from 3.00 V to 5.00 V in steps of 0.25 V. At each input voltage, the output voltage V_{OUT} and the MOSFET gate voltage V_G were measured using a multimeter.

5.2 Observations

The measured output and gate voltages for different input voltages are recorded.

Table 2: Line regulation measurements at $R_L = 22.4\,\Omega$

V_{IN} (V)	V_{OUT} (V)	V_G (V)
3.00	1.2049	4.294
3.25	1.205	4.290
3.50	1.205	4.287
3.75	1.205	4.285
4.00	1.205	4.280
4.25	1.205	4.280
4.50	1.205	4.278
4.75	1.205	4.277
5.00	1.205	4.274

5.3 Graphs

The following graphs illustrate the variation of output voltage and MOSFET gate voltage with input voltage during the line regulation experiment at $R_L = 22.4\,\Omega$.

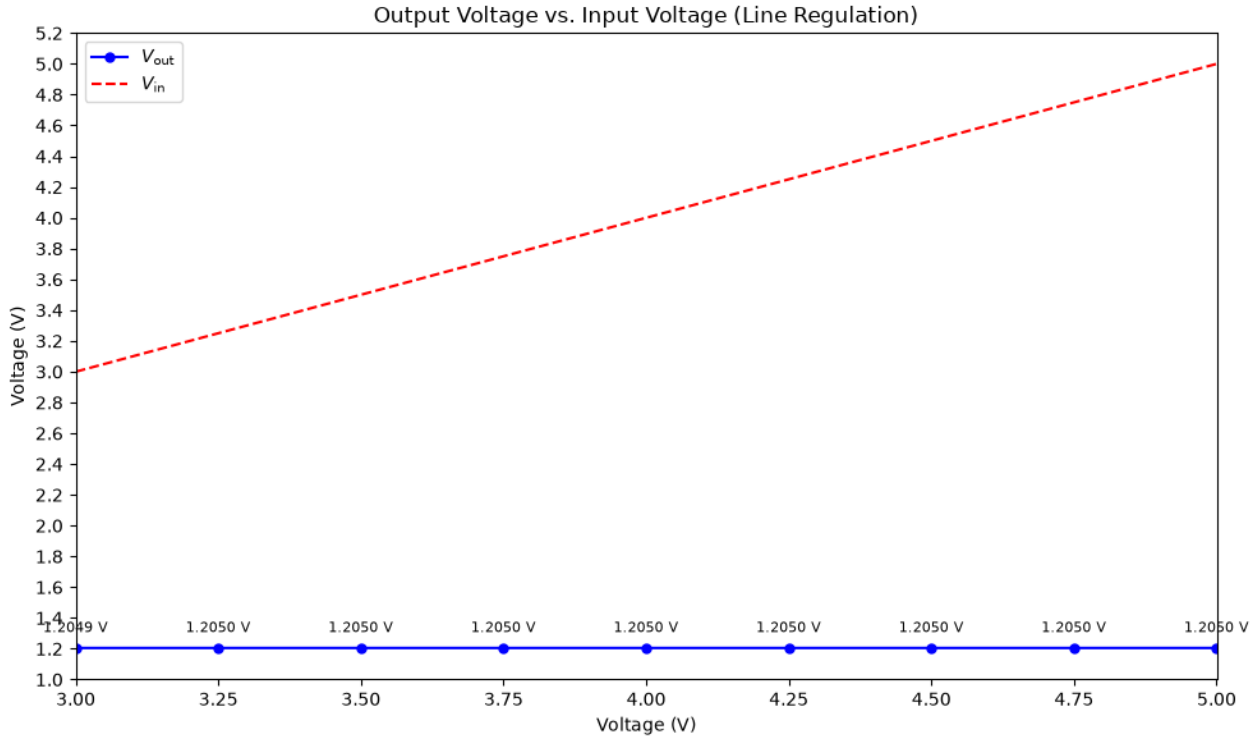


Figure 2: Experimental Plot (V_{out} vs V_{in})

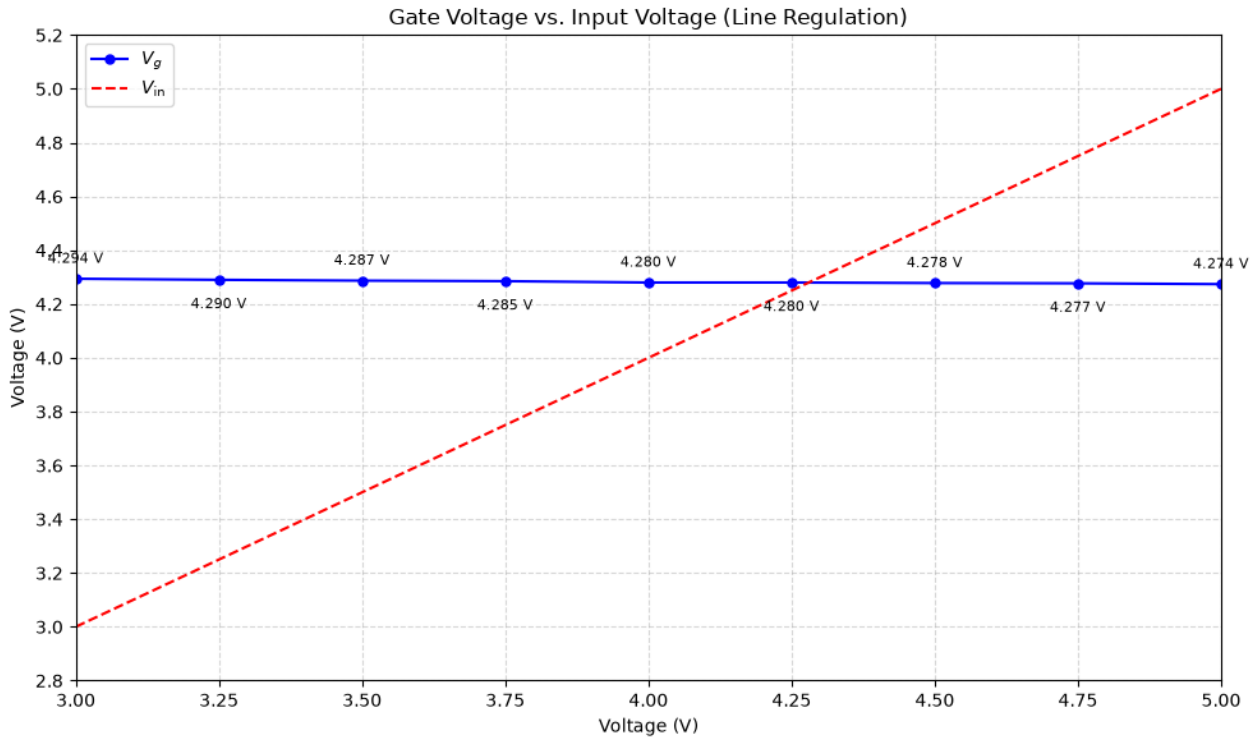


Figure 3: Experimental Plot (V_g vs V_{in})

5.4 Simulations

The circuit was simulated in LTspice by performing a DC sweep of the input voltage from 3 V to 5 V in increments of 0.25 V. The output voltage and MOSFET gate voltage were monitored during the sweep.

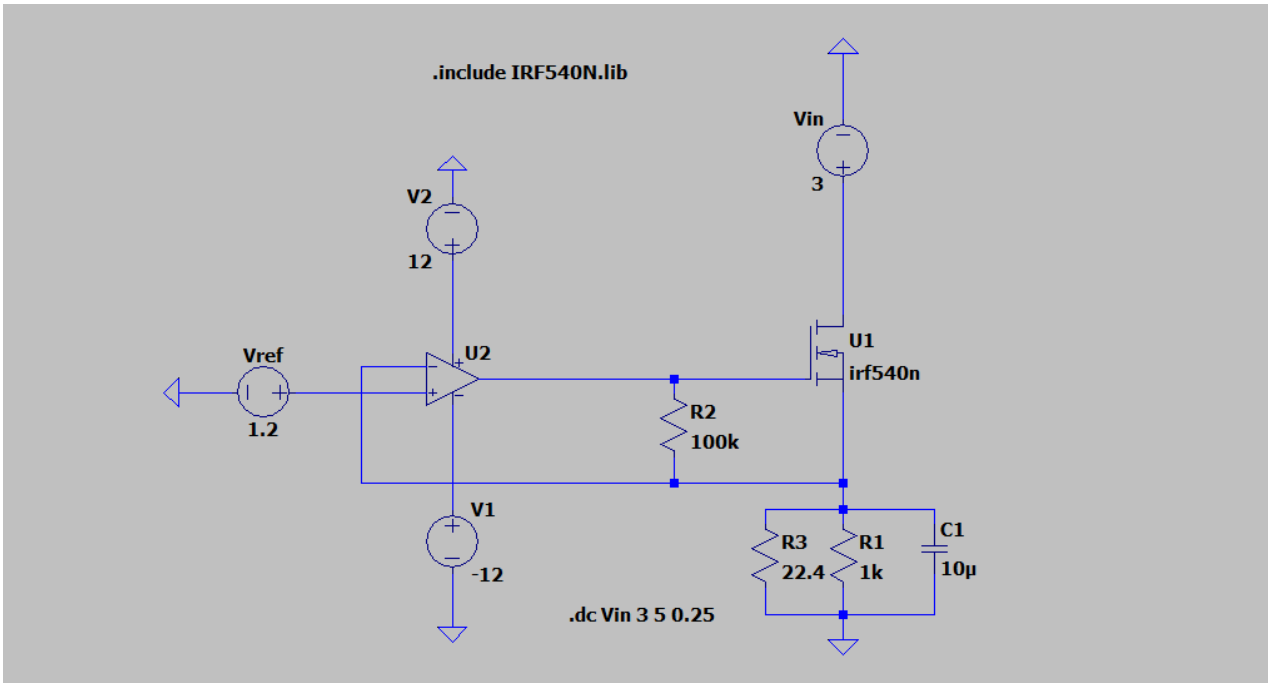


Figure 4: Circuit Schematic

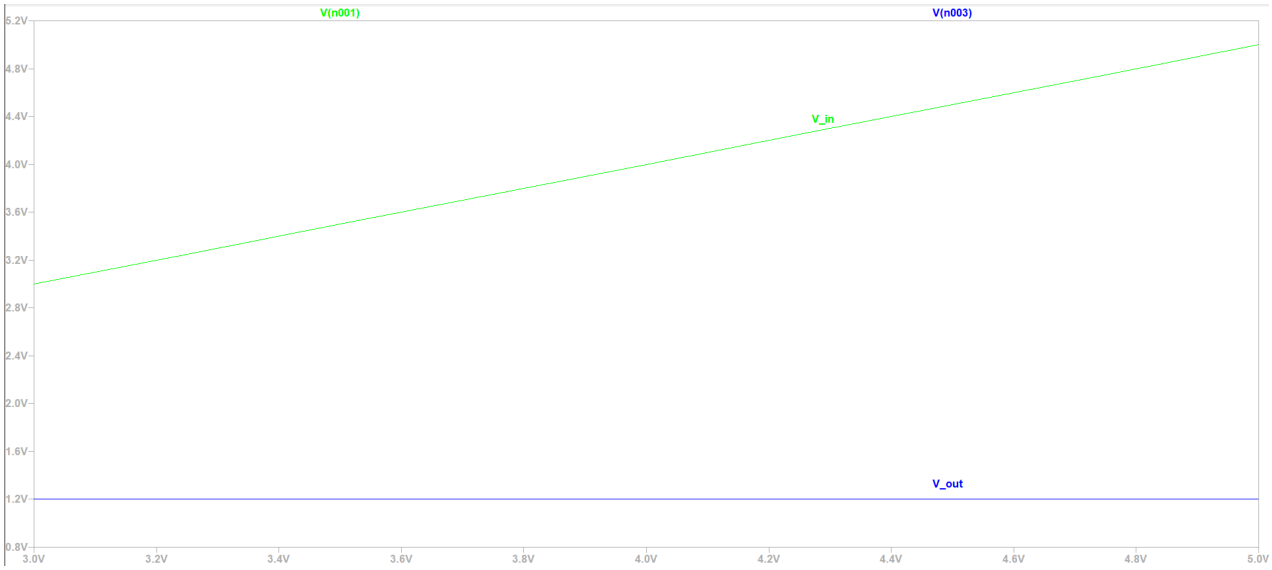


Figure 5: Simulation Plot (V_{out} vs V_{in})

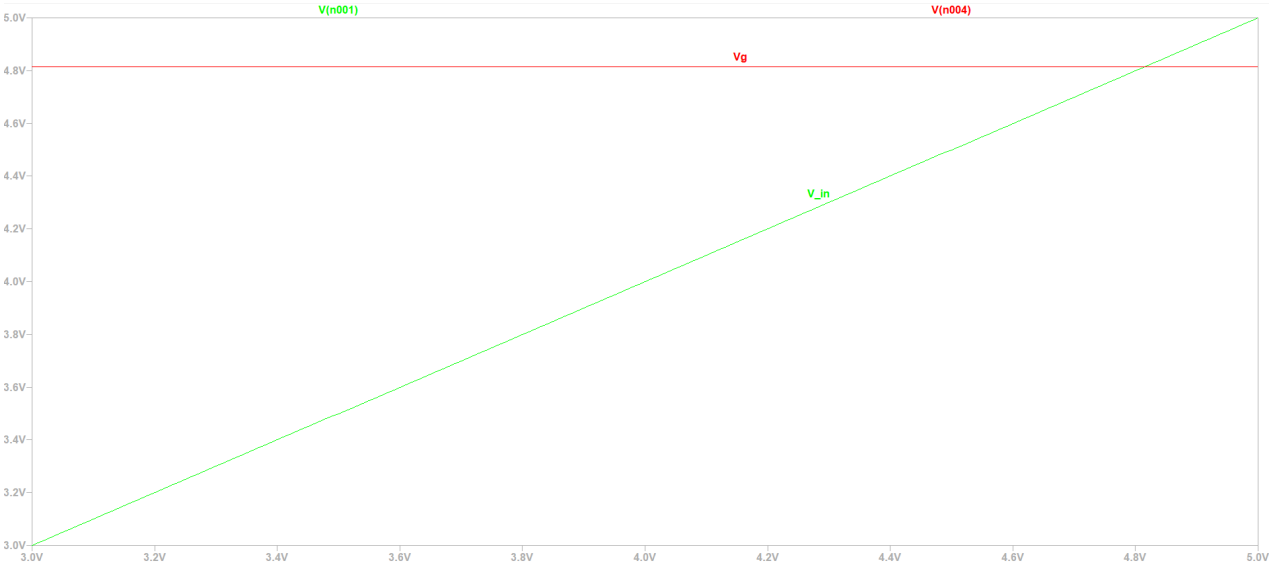


Figure 6: Simulation Plot (V_g vs V_{in})

5.5 Calculations

The line regulation is calculated using the maximum and minimum measured output voltages over the input voltage range.

From the Observations,

$$V_{\mathrm{OUT,max}} = 1.205\,\mathrm{V}$$

$$V_{\mathrm{OUT,min}} = 1.2049\,\mathrm{V}$$

The input voltage variation is

$$\Delta V_{\text{IN}} = 5.00 - 3.00 = 2.00\,\text{V}$$

The change in output voltage is

$$\Delta V_{\text{OUT}} = 1.205 - 1.2049 = 0.0001\,\text{V}$$

Therefore, the line regulation is

$$\begin{aligned}\text{Line Regulation} &= \frac{\Delta V_{\text{OUT}}}{\Delta V_{\text{IN}}} \\ &= \frac{0.0001}{2.00} \\ &= 0.00005\,\text{V/V} \\ &= 0.05\,\text{mV/V}\end{aligned}$$

$$\textbf{Line Regulation} = 0.05\,\text{mV/V}$$

5.6 Results and Discussion

Hence, the measured line regulation was

$$\textbf{Line Regulation} = \textbf{0.05 mV/V}.$$

The output voltage remained approximately constant at 1.205 V over the tested input-voltage range. The measured MOSFET gate voltage decreased from 4.294 V to 4.274 V as the input voltage increased, indicating adjustment of the gate voltage to maintain the regulated output.

The LTspice simulation also showed an approximately constant output voltage of 1.2 V over the input-voltage sweep. Thus, both the experimental and simulation results demonstrate the line regulation behaviour of the circuit. The small measured output variation of 0.1 mV should be interpreted in uncertainty of the measuring instrument.

6 Part C - Load Regulation

6.1 Experimental Procedure

At $V_{IN} = 3\,\text{V}$ and again at 5 V, the load was stepped through the bleeder only and then 124.9, 46.8, 22.4 and 12.2 Ω (measured values) in parallel with the 1 kΩ bleeder. At each step V_{OUT} was measured at both input voltages and V_G at $V_{IN} = 3\,\text{V}$.
The load current was calculated as $I_{LOAD} = V_{OUT}/R_L + V_{OUT}/1\,\text{k}\Omega$.

6.2 Observations

Table 3: Load regulation measurements

R_L (Ω)	I_{LOAD} (mA)	V_{OUT} (V) at $V_{in} = 3V$	V_{OUT} (V) at $V_{in} = 5V$	V_G (V) at $V_{in} = 3V$
Bleeder only	1.19	1.1945	1.1944	3.908
124.9	10.77	1.1960	1.1960	4.127
46.8	26.81	1.1988	1.1995	4.200
22.4	55.00	1.2050	1.2050	4.294
12.2	100.22	1.2080	1.2080	4.320

6.3 Graphs

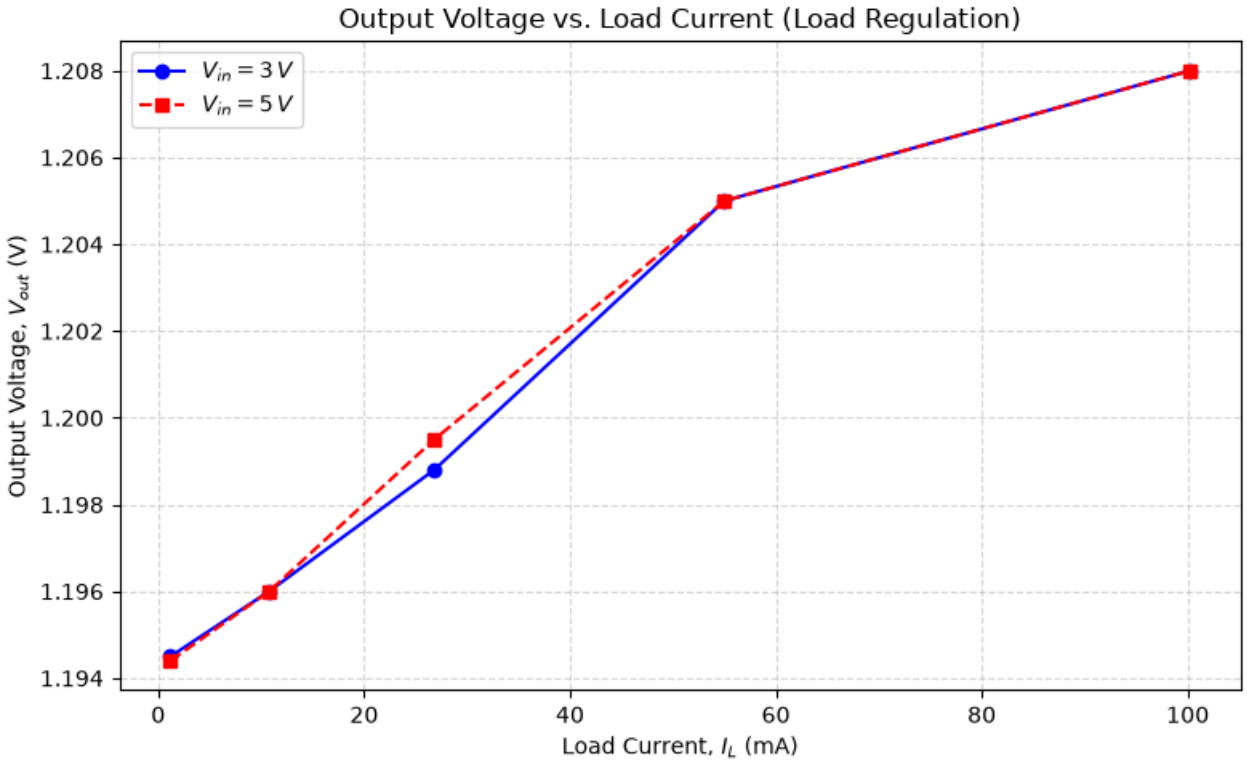


Figure 7: Variation of output voltage with load current at input voltages of 3 V and 5 V.

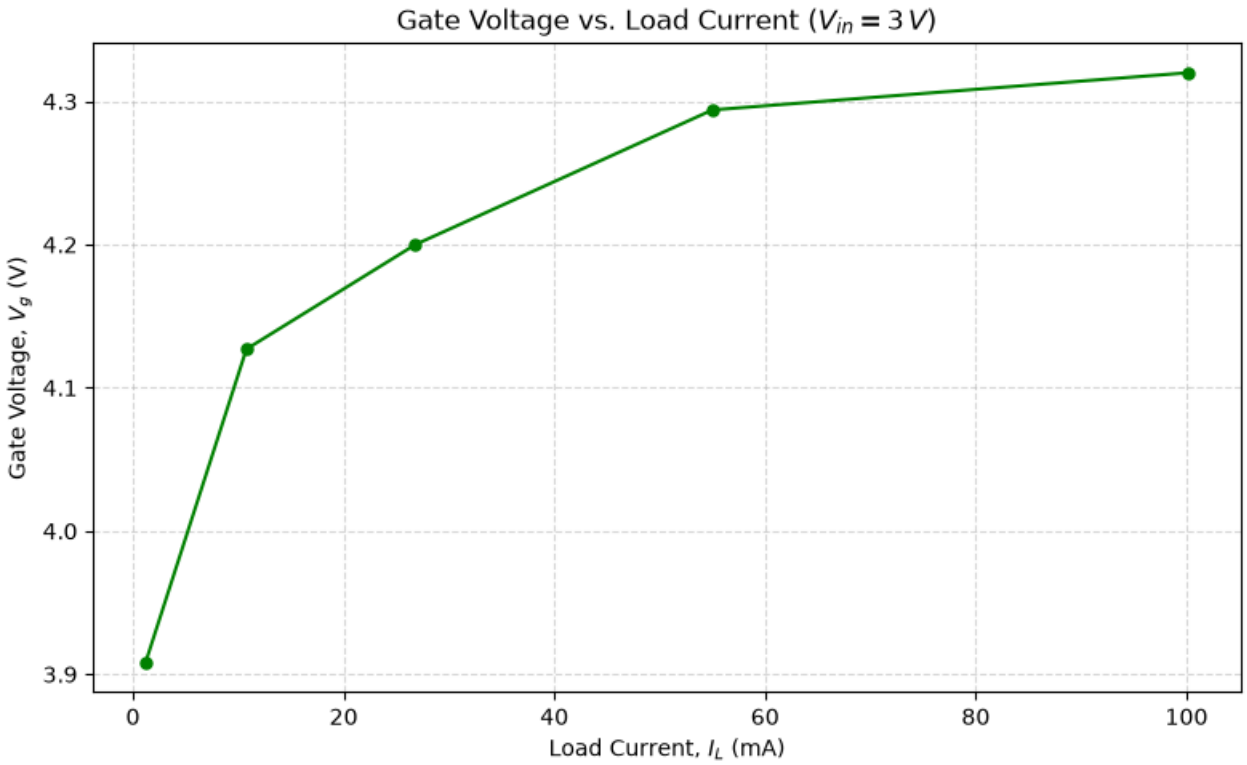


Figure 8: Variation of MOSFET gate voltage with load current at $V_{\text{in}} = 3\,\text{V}$.

6.4 Simulations

In the simulation, the load was modelled by a DC current source I_1 connected in parallel with the $1\,\text{k}\Omega$ bleeder at the output. Sweeping I_1 from 0 to 100 mA gives a continuous load-regulation curve in a single run, and is equivalent to switching the load resistors R_L in parallel with the bleeder in the experiment, since V_{OUT} is held nearly constant by the feedback. $I_1 = 0$ corresponds to the bleeder-only case, and the bleeder adds about 1.2 mA to the plotted external load current.

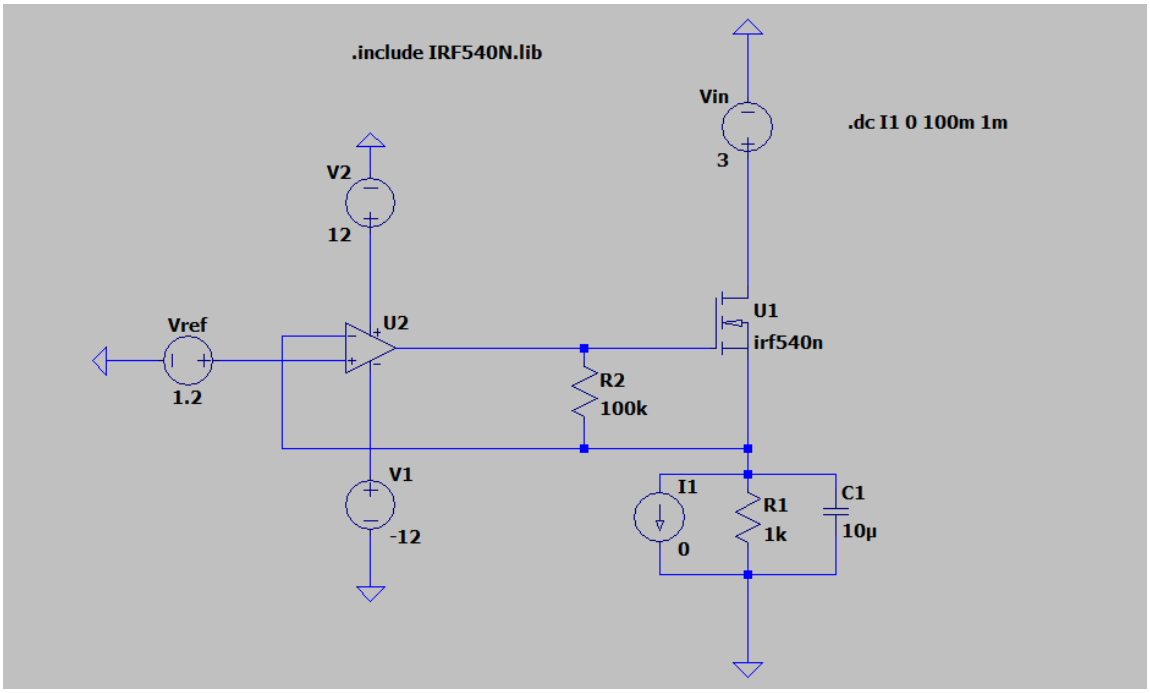
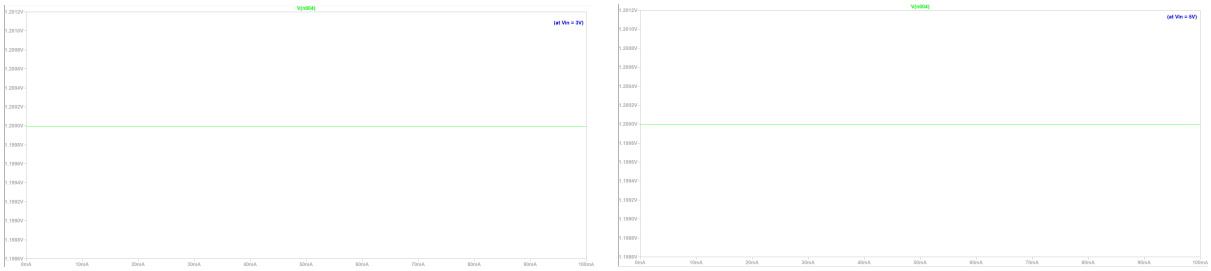


Figure 9: Circuit Schematic



(a) at $V_{in} = 3V$ **(b)** at $V_{in} = 5V$

Figure 10: Simulated plots (V_{out} vs I_{load})

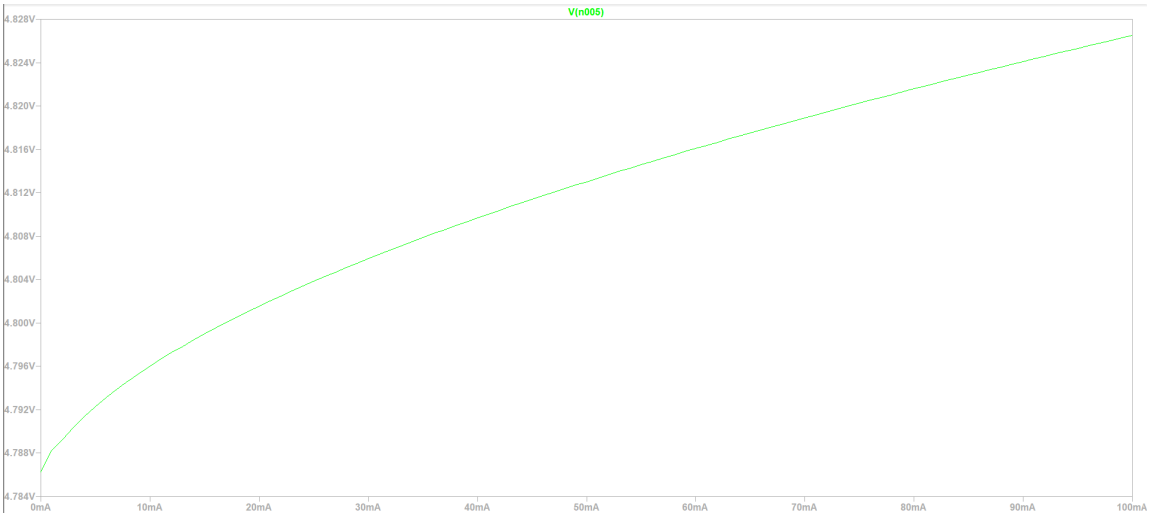


Figure 11: Simulated plot (V_g vs I_{load} at $V_{in} = 3V$)

6.5 Calculations

The load regulation is calculated as the change in output voltage divided by the change in load current.

At $V_{\text{in}} = 3\,\text{V}$:

$$\Delta V_{\text{out}} = 1.2080 - 1.1945 = 0.0135\,\text{V}$$

$$\Delta I_L = (100.22 - 1.19) \times 10^{-3} = 0.09903\,\text{A}$$

$$\begin{aligned}\text{Load Regulation} &= \frac{|\Delta V_{\text{out}}|}{|\Delta I_L|} \\ &= \frac{0.0135}{0.09903} \\ &= 136.32\,\text{mV/A}\end{aligned}$$

At $V_{\text{in}} = 5\,\text{V}$:

$$\Delta V_{\text{out}} = 1.2080 - 1.1944 = 0.0136\,\text{V}$$

$$\begin{aligned}\text{Load Regulation} &= \frac{0.0136}{0.09903} \\ &= 137.33\,\text{mV/A}\end{aligned}$$

6.6 Results and Discussion

The load regulation of the NMOS linear voltage regulator was investigated by varying the load resistance while maintaining the input voltage at 3 V and 5 V.

At $V_{\text{in}} = 3\,\text{V}$, the output voltage increased from 1.1945 V at a load current of 1.19 mA to 1.2080 V at 100.22 mA. The calculated load regulation was 136.32 mV/A.

At $V_{\text{in}} = 5\,\text{V}$, the output voltage increased from 1.1944 V to 1.2080 V over the same load-current range. The calculated load regulation was 137.33 mV/A.

The gate voltage increased from 3.908 V to 4.320 V as the load current increased at $V_{\text{in}} = 3\,\text{V}$. This indicates that the feedback circuit adjusted the MOSFET gate voltage to supply the increased load current.

The results show a small increase in output voltage with increasing load current. The similar load-regulation values at the two input voltages indicate comparable load-dependent behaviour under both operating conditions.

7 Part D: Dropout Voltage

7.1 Experimental Procedure

The dropout characteristics of the NMOS linear voltage regulator were investigated by connecting a load resistance of approximately 12.2 Ω to the output. The input voltage was initially set to 3.0 V and gradually reduced in steps of 0.2 V down to 0 V. The output voltage was measured at every input-voltage setting to determine the point at which the regulator could no longer maintain its regulated output.

7.2 Observations

Table 4: Measured dropout characteristics for $R_L = 12.2\,\Omega$.

V_{IN} (V)	V_{OUT} (V)
3.0	1.210
2.8	1.210
2.6	1.210
2.4	1.210
2.2	1.210
2.0	1.210
1.8	1.210
1.6	1.210
1.4	1.210
1.2	1.192
1.0	0.993
0.8	0.794
0.6	0.596
0.4	0.377
0.2	0.196
0.0	0.0022

7.3 Graphs

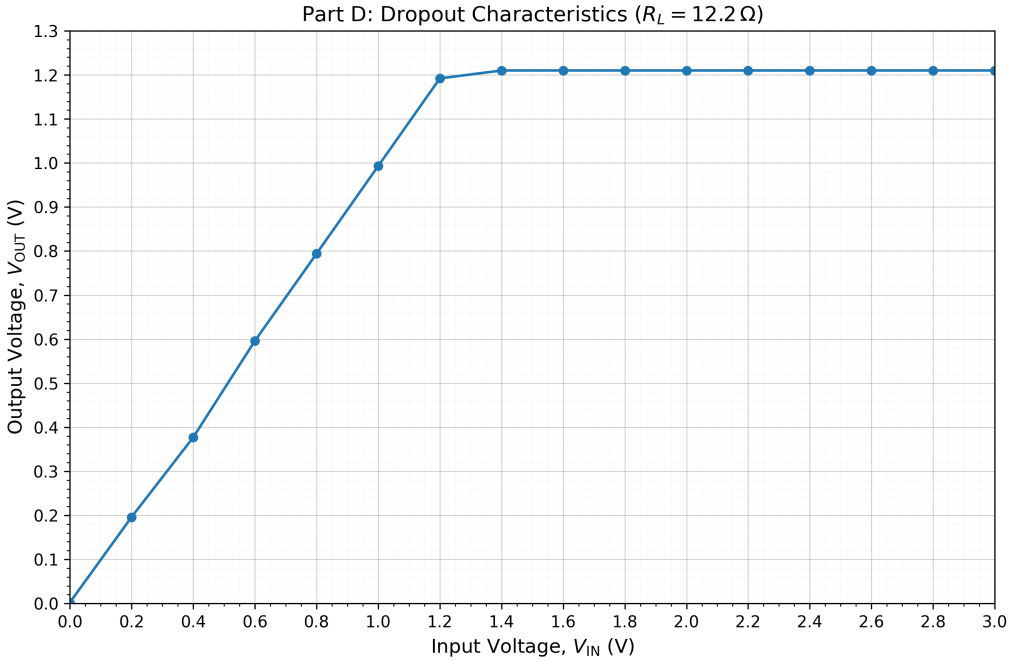


Figure 12: Dropout Voltage Characteristic ($R_L = 12.2\ \Omega$, $I_{LOAD} \approx 100$ mA)

7.4 Simualtions

For the dropout simulation, the same regulator circuit was used with the load fixed at $R_L = 12.2\,\Omega$ in parallel with the $1\,\text{k}\Omega$ bleeder and the $10\,\mu$F output capacitor. The input source V_{IN} was swept from (`.dc Vin 0 3 0.0001`), and V_{OUT} was plotted against V_{IN}.

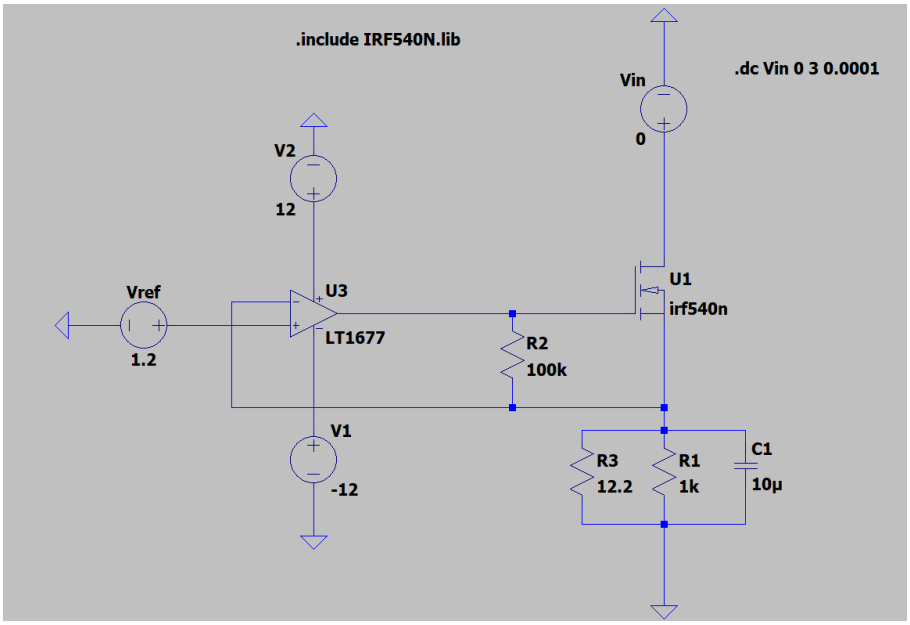


Figure 13: Circuit Schematic

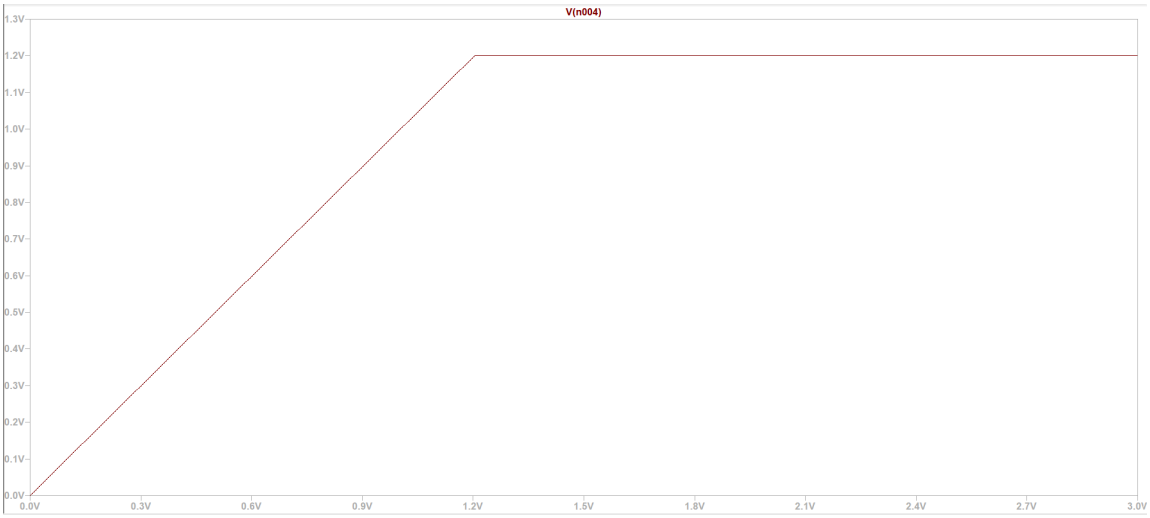


Figure 14: Simualted plot (V_{out} vs V_{in})

7.5 Calculations

The dropout threshold is defined as the input voltage at which the output voltage falls by 1% below its regulated value.

The measured regulated output voltage is

$$V_{\mathrm{OUT,reg}} = 1.210\,\mathrm{V}.$$

The output voltage corresponding to a 1% decrease is

$$V_{\mathrm{OUT,th}} = 0.99 V_{\mathrm{OUT,reg}}.$$

Therefore,

$$V_{\mathrm{OUT,th}} = 0.99(1.210) = 1.1979\,\mathrm{V}.$$

From the observations, the threshold lies between $V_{\mathrm{IN}} = 1.2\,\mathrm{V}$ and $V_{\mathrm{IN}} = 1.4\,\mathrm{V}$. Linear interpolation gives

$$V_{\mathrm{IN,drop}} = 1.2 + \frac{1.1979 - 1.192}{1.210 - 1.192}(1.4 - 1.2).$$

Hence,

$$\boxed{V_{\mathrm{IN,drop}} \approx 1.266\,\mathrm{V}}$$

7.6 Results and Discussion

The output voltage remained constant at approximately 1.210 V as the input voltage decreased from 3.0 V to 1.4 V. This demonstrates the voltage-regulation capability of the circuit.

When the input voltage was reduced further, the output voltage started to decrease. The interpolated input voltage at which the output falls by 1% is approximately 1.266 V.

This behaviour occurs because the regulator eventually loses the ability to maintain the required output voltage when the input supply becomes insufficient. The available voltage across the pass MOSFET decreases, and the op-amp can no longer compensate for the reduction in input voltage.

The estimated dropout threshold is based on interpolation between two measured points and should therefore be considered an approximate experimental result.

8 Part E: Efficiency

8.1 Experimental Procedure

The efficiency of the NMOS linear voltage regulator was investigated using a nominal load resistance of 12 Ω. The input voltage was set successively to 3.0 V, 3.5 V, 4.0 V, 4.5 V, and 5.0 V.

At each input voltage, the input current drawn from the input supply and the output voltage were recorded. The load current was estimated using the output voltage and load resistance.

The input power, output power, pass-stage efficiency, and power dissipation were then calculated. The currents drawn from the auxiliary +12 V and −12 V supplies were also measured to estimate the additional power consumed by the control circuit.

8.2 Observations

Table 5: Measured efficiency characteristics

V_{IN} (V)	I_{IN} (mA)	V_{OUT} (V)
3.0	90.20	1.200
3.5	87.75	1.199
4.0	93.00	1.199
4.5	97.30	1.200
5.0	104.80	1.200

The auxiliary supply current measurements were

$$I_{\mathrm{AUX+}} = 0.35\,\mathrm{mA}$$

$$I_{\mathrm{AUX-}} = -0.35\,\mathrm{mA}.$$

The negative sign indicates the recorded direction of the current through the negative auxiliary supply.

8.3 Calculations

1. Load current

The nominal load resistance is $R_L = 12\,\Omega$. The load current is estimated as

$$I_L = \frac{V_{\mathrm{OUT}}}{R_L}.$$

At $V_{\mathrm{OUT}} = 1.200\,\mathrm{V}$,

$$I_L = \frac{1.200}{12} = 0.100\,\mathrm{A}.$$

Therefore,

$$\boxed{I_L = 100\,\mathrm{mA}}$$

2. Input power

The input power is calculated using

$$P_{\mathrm{IN}} = V_{\mathrm{IN}} I_{\mathrm{IN}}.$$

For example, at $V_{\mathrm{IN}} = 3.0\,\mathrm{V}$,

$$P_{\mathrm{IN}} = 3.0 \times 90.20\,\mathrm{mA} = 270.60\,\mathrm{mW}.$$

3. Output power

The output power delivered to the external load is

$$P_{\mathrm{OUT}} = V_{\mathrm{OUT}} I_L.$$

At $V_{\mathrm{OUT}} = 1.200\,\mathrm{V}$,

$$P_{\mathrm{OUT}} = 1.200 \times 100\,\mathrm{mA} = 120.00\,\mathrm{mW}.$$

4. Pass-stage efficiency

The efficiency is calculated as

$$\eta_{\text{pass}} = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\%.$$

At $V_{\text{IN}} = 3.0\,\text{V}$,

$$\eta_{\text{pass}} = \frac{120.00}{270.60} \times 100 = 44.35\%.$$

5. Power dissipation

Following the experiment handout, the power difference is calculated as

$$P_{\text{diss}} = P_{\text{IN}} - P_{\text{OUT}}.$$

At $V_{\text{IN}} = 3.0\,\text{V}$,

$$P_{\text{diss}} = 270.60 - 120.00 = 150.60\,\text{mW}.$$

6. Calculated results

The calculations for all five input voltages are summarized below.

Table 6: Calculated power and pass-stage efficiency

V_{IN} (V)	P_{IN} (mW)	P_{OUT} (mW)	η_{pass} (%)	P_{diss} (mW)
3.0	270.60	120.00	44.35	150.60
3.5	307.13	119.90	39.05	187.23
4.0	372.00	119.90	32.23	252.10
4.5	437.85	120.00	27.41	317.85
5.0	524.00	120.00	22.90	404.00

7. Auxiliary supply power

The power consumed by the two auxiliary supplies is calculated using the magnitudes of their currents:

$$P_{\text{AUX}} = 12\left(|I_{\text{AUX}+}| + |I_{\text{AUX}-}|\right).$$

Substituting the measured values,

$$P_{\text{AUX}} = 12(0.35 + 0.35)\,\text{mW}.$$

Therefore,

$$\boxed{P_{\text{AUX}} = 8.4\,\text{mW}}$$

8. Total efficiency

The total efficiency, including the auxiliary supply power, is calculated as

$$\eta_{\text{total}} = \frac{P_{\text{OUT}}}{P_{\text{IN}} + P_{\text{AUX}}} \times 100\%.$$

At $V_{\text{IN}} = 3.0\,\text{V}$,

$$\eta_{\text{total}} = \frac{120.00}{270.60 + 8.4} \times 100 \approx 43.01\%.$$

At $V_{\text{IN}} = 5.0\,\text{V}$,

$$\eta_{\text{total}} = \frac{120.00}{524.00 + 8.4} \times 100 \approx 22.54\%.$$

These total-efficiency values use the recorded input currents and the assumed 100 mA external load current. Their physical accuracy is subject to the input-current discrepancy discussed below.

8.4 Graphs

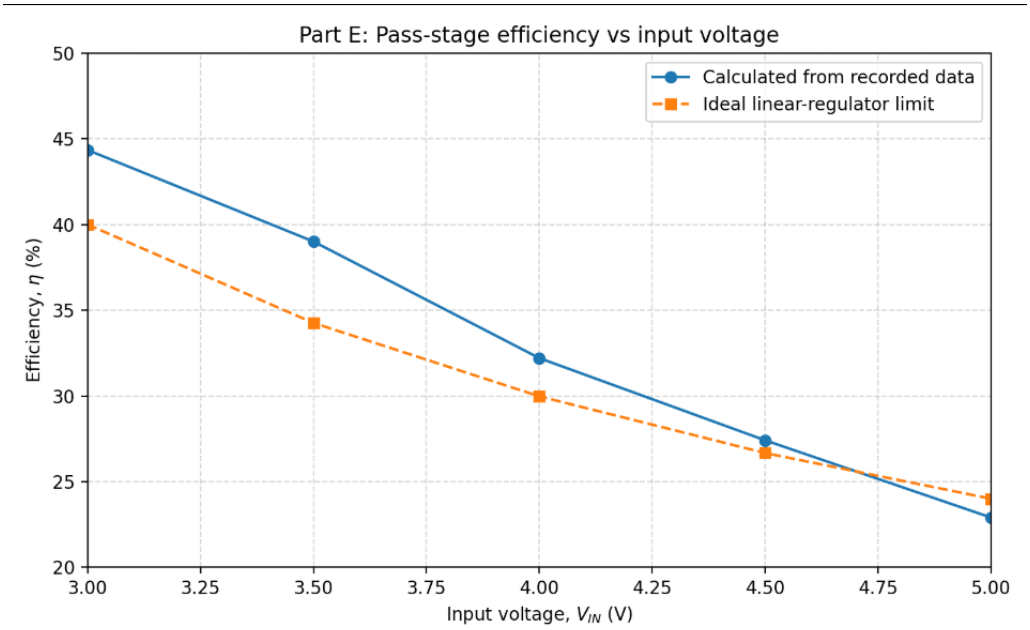


Figure 15: Efficiency vs input voltage

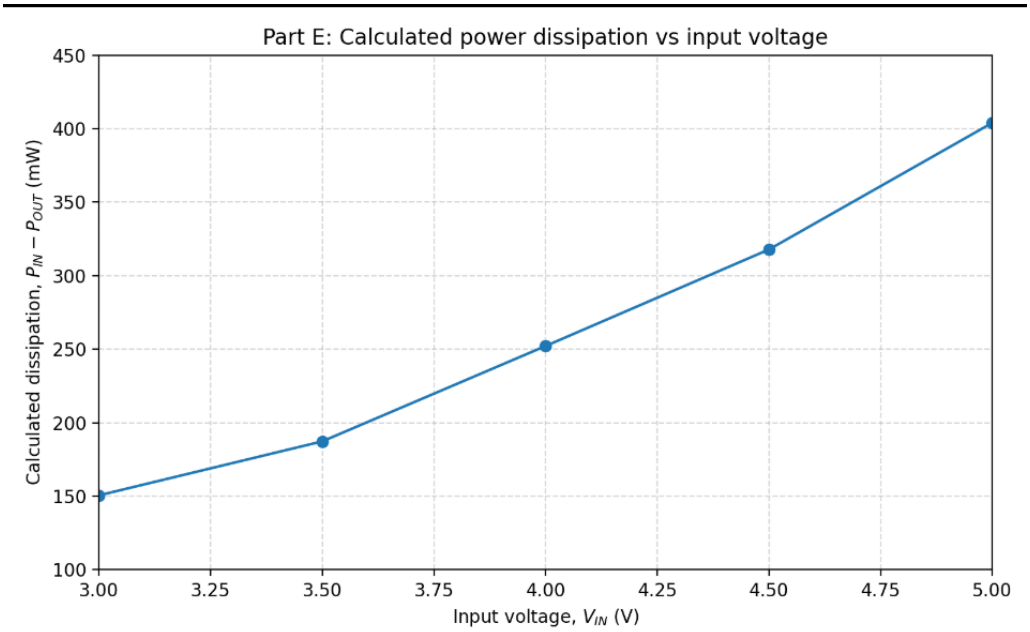


Figure 16: dissipation vs input volatge

8.5 Simulations

For the efficiency simulation, the same regulator circuit as in Part-D was used, with the load fixed at $R_L = 12.2\,\Omega$ in parallel with the $1\,\text{k}\Omega$ bleeder. The input voltage was swept from 3 V to 5 V in 50 mV steps (`.dc Vin 3 5 0.05`).

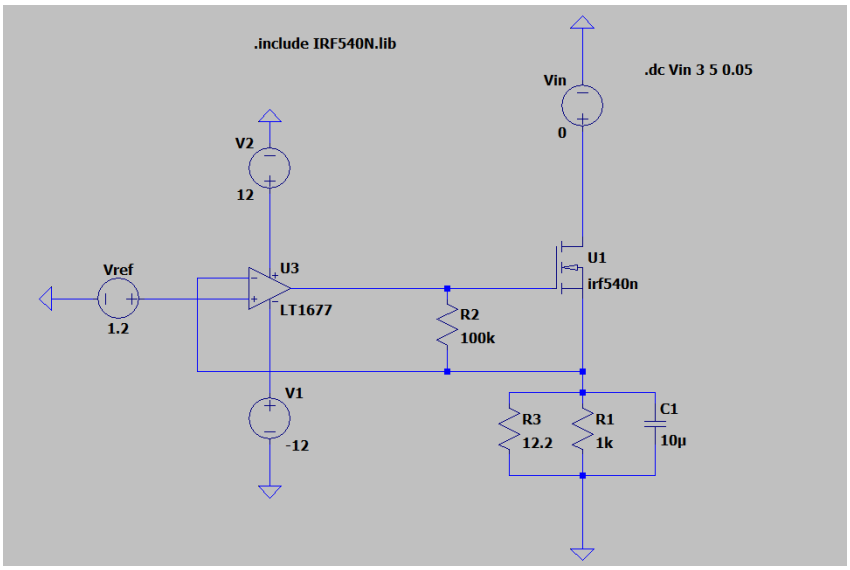


Figure 17: Circuit Schematic

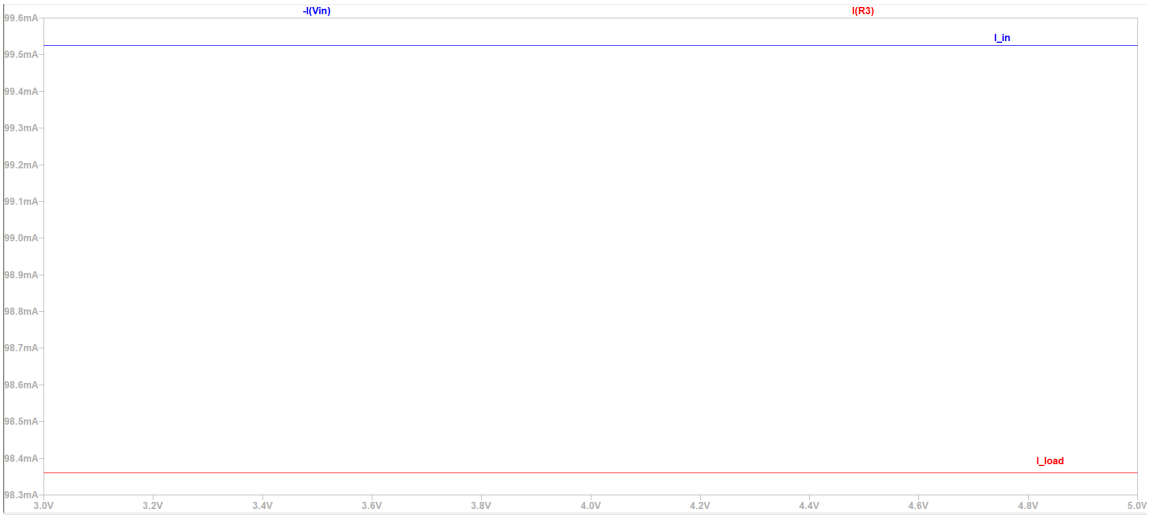


Figure 18: Simulated plot (I_{in} , I_{load} vs V_{in})

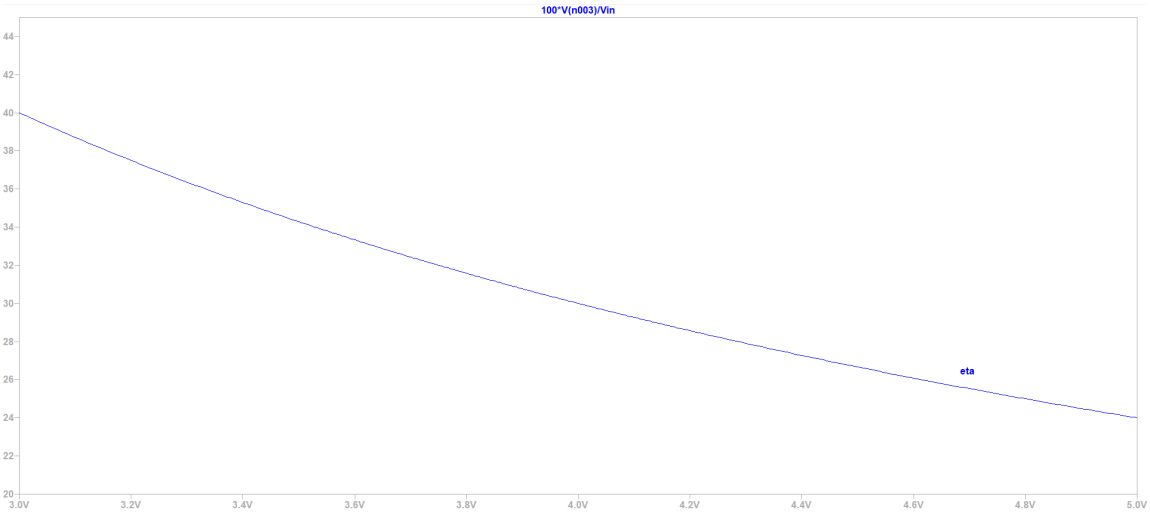


Figure 19: Simulated plot (η vs V_{in})

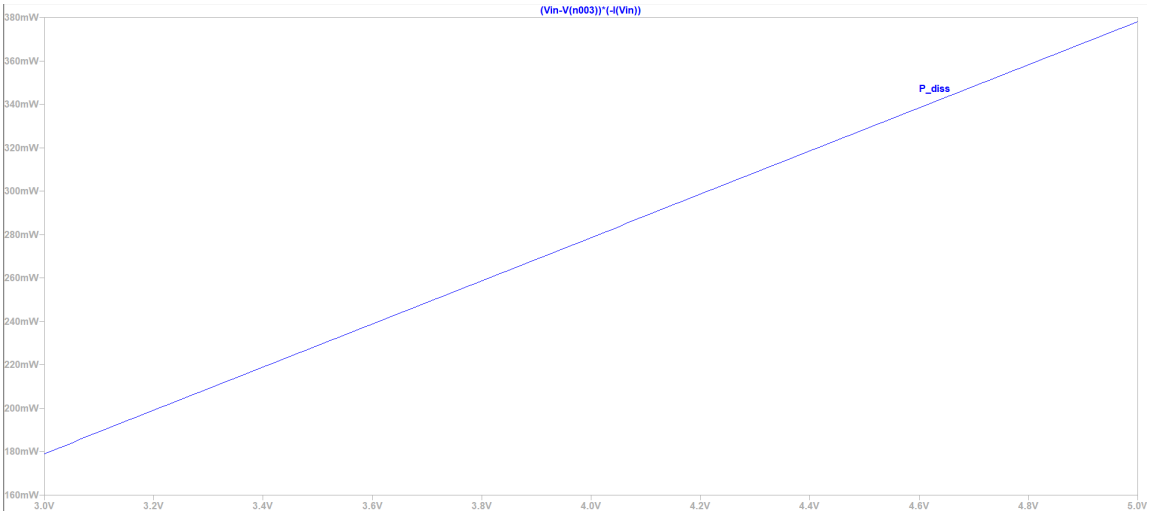


Figure 20: Simulated plot (P_{diss} vs V_{in})

8.6 Results and Discussion

The output voltage remained approximately constant at 1.2 V as the input voltage increased from 3.0 V to 5.0 V. The calculated input power increased from 270.60 mW to 524.00 mW, whereas the output power remained approximately 120 mW.

The calculated pass-stage efficiency decreased from 44.35% at 3.0 V to 22.90% at 5.0 V. The estimated power difference increased from 150.60 mW to 404.00 mW.

This trend is consistent with the general behaviour of a linear regulator: increasing the input voltage while maintaining a constant output voltage increases the power that must be dissipated in the pass transistor.

Explanation of the Part E discrepancy

The calculated efficiencies at 3.0 V and 3.5 V exceed the ideal voltage-ratio limits of approximately 40% and 34.3%, respectively. This indicates that the measured input currents and the assumed output load current are not mutually consistent.

For a linear regulator, the input current supplied through the pass transistor should approximately equal the total current delivered at the output. The circuit also contains a permanent 1 kΩ bleeder resistor in parallel with the external load.

At an output voltage of 1.2 V, the bleeder current is

$$I_B = \frac{1.2}{1000} = 1.2\,\text{mA}.$$

For the nominal 12 Ω external load, the expected total output current is therefore approximately

$$I_{\text{OUT,total}} = I_L + I_B = 100 + 1.2 = 101.2\,\text{mA}.$$

However, the recorded input currents at 3.0 V, 3.5 V, 4.0 V, and 4.5 V are 90.20 mA, 87.75 mA, 93.00 mA, and 97.30 mA, respectively. These are lower than the expected output current of approximately 101.2 mA.

At 5.0 V, the recorded input current is 104.80 mA, which is consistent with the expected current in order of magnitude.

The discrepancy at the lower input voltages may result from an input-current measurement or recording error, an incorrect current-meter connection, a difference between the actual and nominal load resistance, or a circuit connection that differs from the intended setup. The available observations do not establish which explanation is responsible.

The external load current should also be calculated using the actual measured resistance rather than its nominal value if that measurement is available. For example, if the Part E load were actually 12.2 Ω, its current at 1.2 V would be approximately

$$I_L = \frac{1.2}{12.2} \approx 98.36\,\text{mA}.$$

Including the bleeder current gives approximately 99.56 mA, which is still greater than the recorded input currents at the first four settings.

Consequently, the discrepancy cannot be explained solely by the difference between 12 Ω and 12.2 Ω. The input-current readings and actual load resistance should be verified before the efficiency results are considered final.

The original observations have been retained without alteration. The calculated efficiencies are therefore reported as preliminary values rather than being adjusted to force agreement with the expected theoretical result.

9 Part F: Stability Check

9.1 Experimental Procedure

The stability of the NMOS linear voltage regulator was investigated by observing its output voltage using a digital oscilloscope. The output was observed with AC coupling to examine the small variations and noise around the DC output voltage. The measurement was performed under the operating conditions of the regulator.

The oscilloscope was operated in Run mode with automatic acquisition. The vertical scale for Channel 1 was set to 50.0 mV/div, and the horizontal time scale was set to 4.00 µs/div. The acquired waveform contained 2000 points, with a displayed sampling rate of 31.2 MS/s.

9.2 Observations

- Channel used: CH1
- Vertical scale: 50.0 mV/div
- Horizontal scale: 4.00 µs/div
- Sampling rate: 31.2 MS/s
- Number of acquired points: 2000
- Acquisition mode: Run, Auto
- Coupling: AC coupling, as used for the stability measurement

The displayed waveform consists of small fluctuations around the central horizontal level, with occasional short-duration spikes. No clearly visible sustained, periodic oscillation is apparent in the displayed trace.

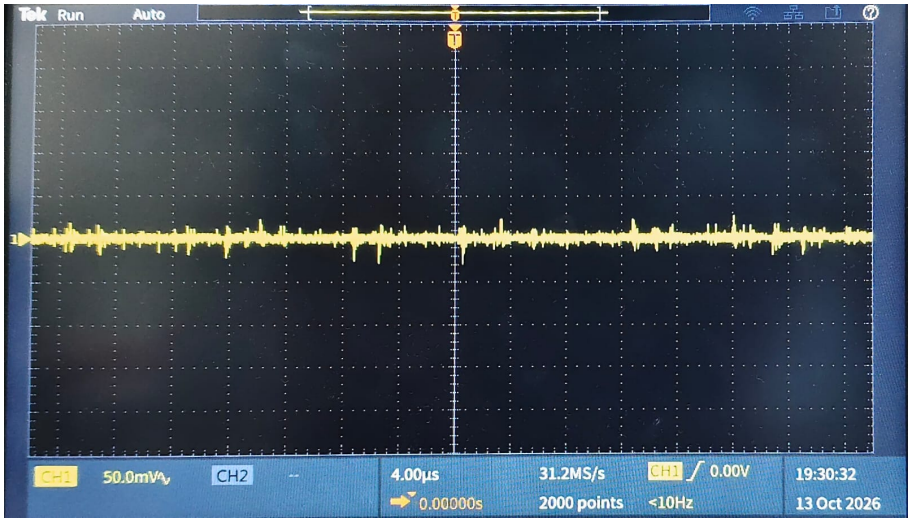


Figure 21: Oscilloscope waveform of the regulator output observed using Channel 1.

9.3 Calculations

The peak-to-peak ripple voltage is defined as

$$V_{\text{ripple,pp}} = V_{\text{OUT,max}} - V_{\text{OUT,min}}.$$

The vertical scale was set to 50.0 mV/div. If the waveform occupies N_{pp} vertical divisions from its minimum to its maximum, the peak-to-peak voltage is

$$V_{\text{ripple,pp}} = N_{\text{pp}} \times 50.0\,\text{mV}.$$

The screenshot does not provide a numerical peak-to-peak measurement, so an accurate value cannot be determined from the displayed scale alone.

If a periodic oscillation is identified, its frequency can be calculated from its measured period:

$$f_{\text{osc}} = \frac{1}{T}.$$

Since no clearly identifiable periodic oscillation or measured period is available, the oscillation frequency is not calculated.

9.4 Results and Discussion

The oscilloscope trace shows small fluctuations around the central horizontal level, together with occasional short-duration spikes. No clearly visible sustained, periodic oscillation is apparent in the displayed waveform.

The absence of an obvious sustained oscillation suggests that the regulator may be stable under the observed conditions. However, the screenshot alone is insufficient to

establish the complete stability of the feedback loop or to determine the exact output-ripple voltage.

The small fluctuations may arise from circuit noise, measurement noise, or other disturbances. The output capacitor helps suppress output-voltage variations, while the negative-feedback loop corrects deviations from the reference voltage.

Therefore, the oscilloscope observation provides a qualitative indication of the regulator's behaviour. A numerical ripple measurement and observations under the specified load conditions would be required for a more complete stability assessment. "'

10 Conclusion

The experiment investigated an NMOS linear voltage regulator controlled by a 741 operational amplifier using negative feedback to maintain an output voltage close to the 1.2 V reference.

In Part A, the measured output voltage was 1.1945 V, and the calculated gate-to-source voltage of 2.7135 V closely matched the measured value of 2.718 V.

In Part B, the output remained approximately 1.205 V for input voltages from 3 V to 5 V, giving a line regulation of 0.05 mV/V. In Part C, the calculated load regulation was 136.32 mV/A at 3 V and 137.33 mV/A at 5 V, demonstrating the regulator's response to changing load current.

In Part D, the output remained near 1.210 V until the input approached the dropout region, with an interpolated 1% dropout threshold of approximately 1.266 V. In Part E, the efficiency decreased as the input voltage increased, while power dissipation increased. However, inconsistencies in the measured input currents require verification before the efficiency results can be considered final.

In Part F, the oscilloscope showed small fluctuations and occasional spikes, with no clearly visible sustained periodic oscillation. Further measurements are needed to quantify output ripple and confirm stability.

Overall, the experiment demonstrated the regulator's line and load regulation, dropout behaviour, and power-efficiency trade-offs, while highlighting the importance of measurement accuracy.